Unit 3 — What if 2026-XJ hits, and can we nudge it?

Momentum · impulse · conservation · collisions · asteroid deflection (DART)

◆ NASA / PLANETARY DEFENSE BRIEFING — UPDATE 3

Unit 2 closed with you computing 2026-XJ's closest-approach date. The number was uncomfortably close. NASA has now run high-precision ephemeris simulations and confirmed: at current trajectory, 2026-XJ and Earth will arrive at the orbit-crossing point within HOURS of each other.

Refined closest-approach distance: somewhere between 40,000 and 100,000 km, depending on remaining orbital uncertainties. For reference, Earth's radius is 6,371 km, and the Moon orbits Earth at about 384,000 km. So: not a definite impact — but uncomfortably INSIDE the Moon's orbit.

The two questions Unit 3 must answer: (1) WHAT IF IT DOES HIT — what does the physics say about the impact? (2) CAN WE NUDGE IT — is there a way to push the asteroid OFF course before it gets here?

Precedent on the table: in 2022, NASA's DART mission deliberately crashed a 600-kg spacecraft into a small asteroid called Dimorphos and changed its orbital period by 32 minutes. The math behind DART is exactly the math you'll learn this unit.

This is fiction. The physics — and the equations — are real.

MA Standards: HS-PS2-2, HS-PS2-3, HS-ETS1-2

14 meetings · Fall 2026 (Nov 9 – Dec 1)

Name: ______________________________	Class period: __________	Date

received: __________

"Unit 2 told us WHERE. Unit 3 asks: what then — and is there anything we can do?"

Table of Contents

Day-type key: A = Anchor · S = Standard · L = Lab · T = Transfer

Day 1 — Re-anchoring — What If It Hits, And Can We Nudge It?

Today's Target

I can name the two things that make a moving object hard to stop, and I can frame the two questions Unit 3 must answer about 2026-XJ.

ASTEROID THREAD

What we know about 2026-XJ today: From Unit 2: 2026-XJ and Earth arrive at the orbit-crossing point within HOURS of each other. Closest-approach distance: 40,000–100,000 km — inside the Moon.

How today connects: Unit 2 told us WHERE 2026-XJ is going. Today we open Unit 3 with two new questions: (1) IF it hits, what does the physics say? (2) Could we PUSH it off course before it gets here, like NASA's DART mission did in 2022?

NOTICE AND WONDER

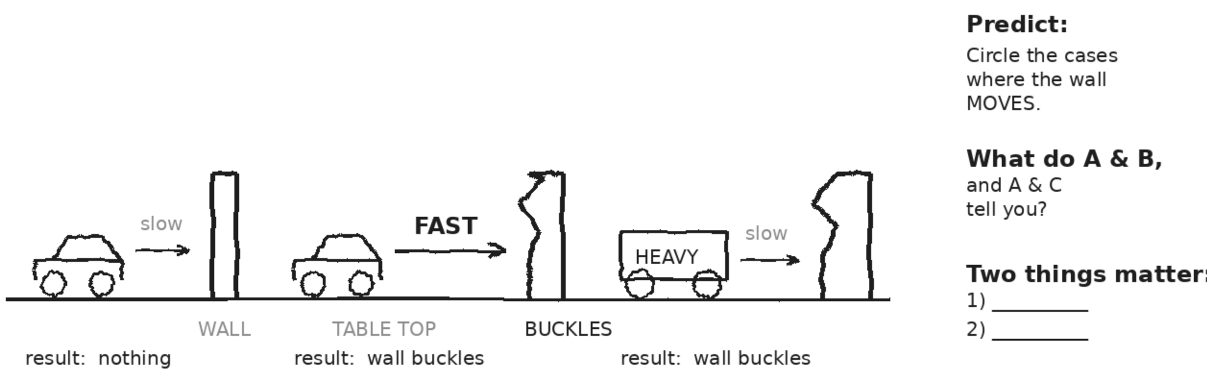
Watch the demo: Hot Wheels car gently into a cardboard wall vs. the SAME car going FAST vs. a HEAVIER cart at the slow speed. What did you NOTICE? What do you WONDER?

I notice ____________________________

I wonder ____________________________

DOODLE ZONE

Circle the cases where the wall MOVES. What do A & B tell you? What do A & C tell you? What TWO things matter?



MY ONE QUESTION ABOUT DEFLECTING (OR SURVIVING) AN ASTEROID

EXIT TICKET — DAY 1
Between (a) a 1500-kg car at 30 m/s and (b) a 0.05-kg bullet at 400 m/s, which would be HARDER to stop with your hand? Why? (Both have honest defenses — name the trade-off.)

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can name TWO properties that make a moving object hard to stop.	○	○	○	☐
I can state the two Unit 3 questions in my own words.	○	○	○	☐

Day 2 — Momentum p = m·v

Today's Target

I can write p = m·v, give the units (kg·m/s), compute momentum for a moving object, and explain why momentum is a VECTOR (sign and direction matter).

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ's mass is ~10^9 kg, its speed (with respect to Earth) is ~30 km/s.

How today connects: Today we put a NUMBER on what makes the asteroid hard to stop. Spoiler: the number is enormous, and the rest of Unit 3 figures out what to do with that fact.

VOCAB · MOMENTUM (P)

Definition: A measure of how hard it is to stop a moving object. p = m·v. Vector — has the same direction as the velocity.

Example: A 1500-kg car at 30 m/s has p = 45,000 kg·m/s. A 0.05-kg bullet at 400 m/s has p = 20 kg·m/s. The car wins.

Cognates: *Sp. cantidad de movimiento (or momento lineal) · Pt. quantidade de movimento · HC kantite mouvman*

Watch out: Momentum is NOT the same as speed. A slow heavy thing can have MORE momentum than a fast light thing.

DOODLE ZONE

Compute p for each row. Then rank from biggest to smallest momentum. Look at the spread.

Five objects · same direction (east). Compute p = m·v for each.

CAR	m = 1500 kg	v = 30 m/s	p = ________ kg·m/s
BULLET	m = 0.05 kg	v = 400 m/s	p = ________ kg·m/s
SPRINTER	m = 70 kg	v = 10 m/s	p = ________ kg·m/s
CRUISE	m = 5×10^6 kg	v = 8 m/s	p = ________ kg·m/s
2026-XJ	m = $\sim10^9$ kg	v = ~30 km/s	p = ________ kg·m/s

Rank the five
by momentum:

1. __________
2. __________
3. __________
4. __________
5. __________

WORKED EXAMPLE — MOMENTUM OF A CAR

A 1500-kg car moves east at 30 m/s. Find its momentum (magnitude AND direction).

GIVEN	m = 1500 kg · v = 30 m/s east
EQUATION	p = m·v
WORK	p = 1500 × 30 = 45,000 kg·m/s p = 4.5×10^4 kg·m/s east
ANSWER	4.5×10^4 kg·m/s east. (East matters — p is a vector.)

YOUR TURN — MOMENTUM OF 2026-XJ

2026-XJ has mass $m = 1.0 \times 10^9$ kg and approaches at $v = 3.0 \times 10^4$ m/s toward Earth. Find its momentum.

GIVEN	
EQUATION	
WORK	
ANSWER	

YOUR TURN — Δp ON A BASEBALL (FORCE THE SIGN)

A 0.15-kg baseball moves east at 40 m/s, meets the bat, and bounces back west at 50 m/s. What is the CHANGE in its momentum, $\Delta p = p_f - p_i$? (Hint: pick east = positive, so $v_f = -50$.)

GIVEN	
EQUATION	
WORK	
ANSWER	

⚠	Sign matters. If you ignore the negative on v_f, your Δp will look like 1.5 kg·m/s. The real Δp is −13.5 kg·m/s — almost 10× bigger. The direction of Δp is opposite v_i.

Fill in the frame:

Momentum is bigger for the cruise ship than the bullet because, even though the ship is ____, its ____ is overwhelmingly larger.

EXIT TICKET — DAY 2
A 0.15-kg baseball moves east at 40 m/s. After the bat, it goes WEST at 50 m/s. Find Δp. Force the sign.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	**1**	**2**	**3**	**Help?**
I can compute $p = m \cdot v$ with correct units.	◯	◯	◯	☐
I can identify momentum as a vector and assign signs to opposite directions.	◯	◯	◯	☐
I can compute Δp using signed velocities.	◯	◯	◯	☐

Day 3 — Impulse: $J = F \cdot \Delta t = \Delta p$

Today's Target

I can derive $F\Delta t = \Delta p$ from $F = ma$, name $F\Delta t$ as impulse, and use the equation to find force, time, or change in momentum given the other two.

ASTEROID THREAD

What we know about 2026-XJ today: To CHANGE 2026-XJ's momentum we need to apply a force OVER A TIME.

How today connects: Today's equation IS the DART equation. A small spacecraft, hitting an asteroid for a tiny instant, delivers a Δp — same math we use for a bat hitting a ball.

VOCAB · IMPULSE (J)

Definition: The product of force and the time it acts: $J = F \cdot \Delta t$. Equals the change in momentum Δp.

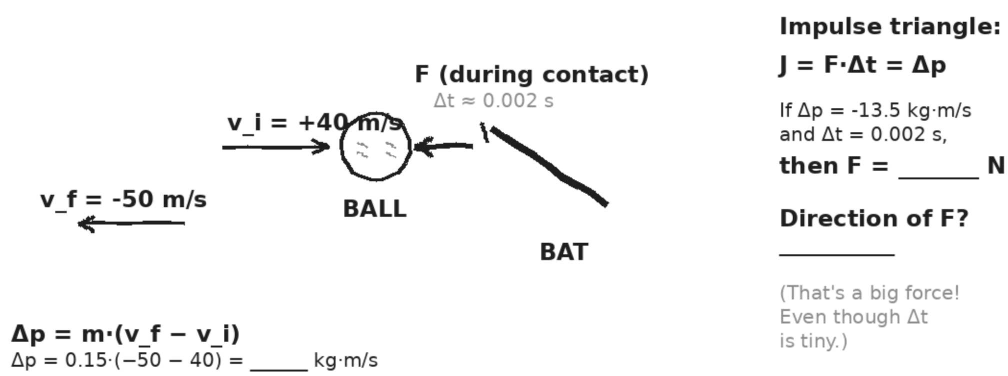
Example: A 0.002-s bat-hit at 6,750 N delivers $J = 13.5$ N·s of impulse to the baseball.

Cognates: *Sp. impulso · Pt. impulso · HC enpilsyon*

Watch out: Impulse is NOT just force. It's force × TIME. Half the force for twice the time gives the same impulse.

DOODLE ZONE

Fill the Δp blank. Then use J = F·Δt to find the force during contact. Bigger than you expected? Why?



DERIVATION — WHERE FΔT = ΔP COMES FROM

Start: $F = m \cdot a = m \cdot (\Delta v / \Delta t)$

Multiply both sides by Δt: $F \cdot \Delta t = m \cdot \Delta v = \Delta p$

Name $F \cdot \Delta t = J$ (impulse). Then: $J = \Delta p$. Units: N·s = kg·m/s.

WORKED EXAMPLE — FORCE DURING A BAT-BALL HIT

A 0.15-kg baseball moves east at 40 m/s, then rebounds WEST at 50 m/s. Bat-ball contact lasts 0.002 s. Find (a) Δp, (b) the average force.

GIVEN	$m = 0.15$ kg · $v_i = +40$ m/s · $v_f = -50$ m/s · $\Delta t = 0.002$ s
EQUATION	$\Delta p = m \cdot (v_f - v_i)$; $F = \Delta p / \Delta t$
WORK	$\Delta p = 0.15 \cdot (-50 - 40) = 0.15 \cdot (-90) = -13.5$ kg·m/s $F = -13.5 / 0.002 = -6750$ N
ANSWER	$\Delta p = -13.5$ kg·m/s (westward). $F \approx -6750$ N (westward, applied BY the bat ON the ball).

YOUR TURN — DART'S PUNCH

DART hit Dimorphos for about 0.05 s with an average force of about 8×10^5 N. What change in asteroid momentum did it deliver?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	The same J can come from BIG F over a SHORT Δt or SMALL F over a LONG Δt. That choice is tomorrow's payoff.

EXIT TICKET — DAY 3
DART hit its target for 0.05 s with an average 8×10^5 N. What Δp did it deliver to the asteroid?

Today's Targets — Rate Yourself

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can derive $F\Delta t = \Delta p$ from $F = ma$.	○	○	○	☐
I can compute impulse given F and Δt.	○	○	○	☐
I can find force given Δp and Δt.	○	○	○	☐

Day 4 — Why Impulse Matters — Airbags, Follow-through, Padding

Today's Target

I can explain why extending Δt reduces the force needed to deliver the same Δp, and I can apply this to airbags, padded floors, follow-through, and asteroid deflection.

ASTEROID THREAD

What we know about 2026-XJ today: To deflect 2026-XJ we want the LARGEST Δp possible — either short-and-hard (DART) or long-and-soft (gravity tractor).

How today connects: Same physics that saves lives in a car crash (long Δt, small F) shapes how planetary defense engineers design asteroid pushers.

VOCAB · CRUMPLE ZONE / CUSHION

Definition: A region designed to DEFORM during impact, extending Δt and so reducing peak force.

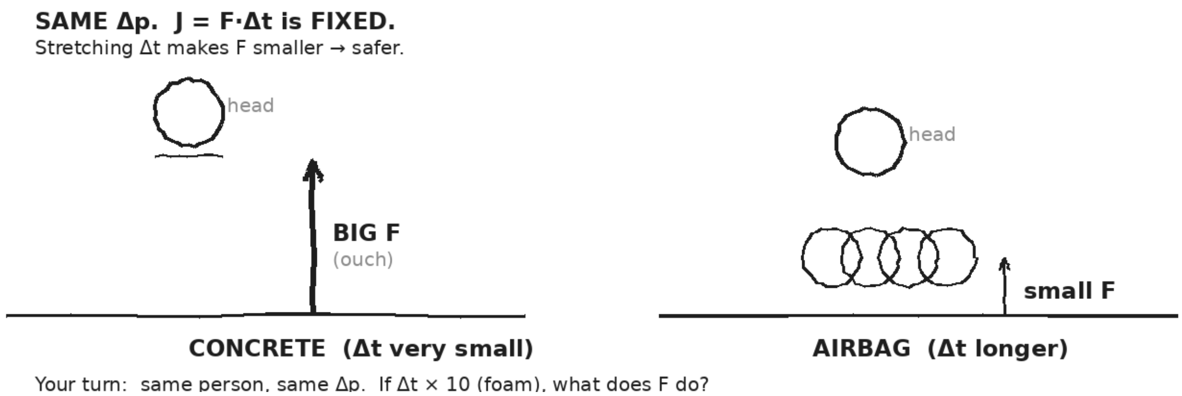
Example: Modern car fronts are designed to crumple. Same Δp, but $\Delta t \times 5$ means $F \div 5$.

Cognates: *Sp. zona de absorción · Pt. zona de deformação · HC zòn dezespere*

Watch out: A crumple zone doesn't 'absorb momentum.' Δp is the same. It SPREADS THE TIME so peak F drops.

DOODLE ZONE

Compare the F arrows in both panels. Same Δp; different Δt. Fill the blank: if Δt × 10, F goes to ____ of original.



WORKED EXAMPLE — SPRINTER INTO CONCRETE VS. INTO FOAM

An 80-kg sprinter at 9 m/s stops in (a) 0.05 s on concrete, or (b) 0.5 s into a foam pit. Compute the average force in each case.

GIVEN	$m = 80$ kg · $\Delta v = 9$ m/s (he stops) · $\Delta t_a = 0.05$ s · $\Delta t_b = 0.5$ s
EQUATION	$\Delta p = m \cdot \Delta v$; $F = \Delta p / \Delta t$
WORK	$\Delta p = 80 \cdot 9 = 720$ kg·m/s (same for both) $F_a = 720 / 0.05 = 14{,}400$ N (concrete) $F_b = 720 / 0.5 = 1{,}440$ N (foam — 10× smaller!)
ANSWER	Concrete: 14.4 kN. Foam: 1.44 kN. Stretching Δt by 10× cut F by 10×.

YOUR TURN — CASE-STUDY REASONING

Pick one: airbag, helmet, boxing glove, gymnastics mat, parkour roll, or follow-through on a tennis swing. Explain how it uses LONGER Δt to reduce F for the SAME Δp. Be specific about what extends the contact time.

GIVEN	
EQUATION	
WORK	
ANSWER	

★	The SAME logic, with a different sign: a baseball hitter wants MORE Δp delivered to the ball. Follow-through extends Δt. Bigger J = bigger Δp = ball goes farther.

Fill in the frame:

A boxer 'rolls with the punch' so that

____ stays roughly the same

but ____ gets larger — which

makes ____ smaller.

EXIT TICKET — DAY 4

Why does a boxer 'roll with the punch'? Use 'Δp', 'Δt', and 'F' in your answer.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can explain why stretching Δt reduces F for the same Δp.	○	○	○	☐
I can apply impulse reasoning to a real-world safety device.	○	○	○	☐
I can predict how a sport technique uses impulse to its advantage.	○	○	○	☐

Day 5 — Conservation of Momentum — Explosions and Recoil

Today's Target

I can state conservation of momentum ($\Sigma p\_before = \Sigma p\_after$) and its condition (no net external force), and I can use it to solve recoil/explosion problems.

ASTEROID THREAD

What we know about 2026-XJ today: When DART hit Dimorphos, total momentum BEFORE = total momentum AFTER. The asteroid sped up; the spacecraft slowed (and was destroyed).

How today connects: Today's rule is the BIG one of the unit. The next eight days are all special cases of it.

VOCAB · CONSERVATION OF MOMENTUM

Definition: If the NET external force on a system is zero, the TOTAL momentum of the system stays constant: $\Sigma p\_before = \Sigma p\_after$.

Example: A rifle at rest fires a bullet east. The bullet goes east; the rifle recoils west. Total p stays zero.

Cognates: *Sp. conservación del momento · Pt. conservação do momento · HC konsèvasyon mouvman*

Watch out: The rule isn't 'no force acts on the objects.' Internal forces (between bullet and rifle) are fine. The rule needs no NET EXTERNAL force.

VOCAB · RECOIL

Definition: When an object at rest splits into two pieces (or expels mass), the pieces move in opposite directions so total p stays the same.

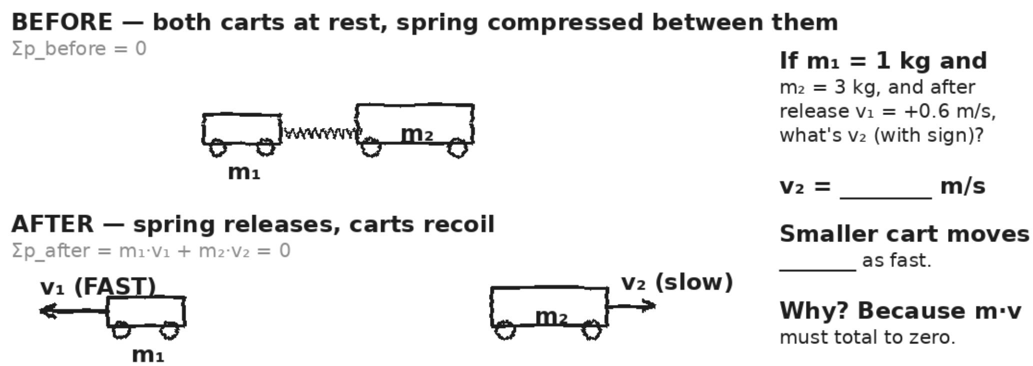
Example: A swimmer pushes a kickboard east; she drifts west. Total p = 0 stays 0.

Cognates: *Sp. retroceso · Pt. recuo · HC rekil*

Watch out: The smaller mass moves FASTER. $m_1v_1 = m_2v_2$ in size, opposite sign. A tiny bullet recoils a heavy gun only a little.

DOODLE ZONE

Use $\Sigma p\_before = 0 = m_1 v_1 + m_2 v_2$ to predict v_2. Smaller cart moves faster (or slower?). Why?



WORKED EXAMPLE — ICE SKATER THROWS A BALL

A 70-kg ice skater (initially at rest) throws a 5-kg ball east at 8 m/s. What is the skater's recoil velocity?

GIVEN	m_skater = 70 kg · m_ball = 5 kg · v_ball = +8 m/s east · both at rest before
EQUATION	Σp_before = Σp_after → 0 = m_skater·v_skater + m_ball·v_ball
WORK	0 = 70·v_skater + 5·(+8) 0 = 70·v_skater + 40 v_skater = −40/70 ≈ −0.57 m/s
ANSWER	Skater drifts WEST at about 0.57 m/s. (Small, because she's 14× heavier than the ball.)

YOUR TURN — FIREWORK EXPLOSION

A 0.5-kg firework at REST explodes into two pieces: a 0.3-kg chunk goes east at 12 m/s. The other piece (0.2 kg) goes — what direction and how fast?

GIVEN	
EQUATION	
WORK	
ANSWER	

⚠	Forces between the pieces during the explosion are INTERNAL — they don't break conservation. Only EXTERNAL forces (like gravity, friction) can change total p.

Fill in the frame:

Total momentum stays the same because the only forces during the explosion are ____ (between the pieces), which means no NET ____ force.

EXIT TICKET — DAY 5
A 0.5-kg firework at rest explodes into a 0.3-kg piece moving east at 12 m/s and a 0.2-kg piece moving — what direction and how fast?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can state Σp_before = Σp_after and its condition.	○	○	○	☐
I can solve recoil problems with signed velocities.	○	○	○	☐
I can identify what counts as INTERNAL vs. EXTERNAL force in a system.	○	○	○	☐

Day 6 — Three Flavors of Collision

Today's Target

I can distinguish elastic, inelastic, and perfectly inelastic collisions, and I can identify which real-world collisions fit each category.

ASTEROID THREAD

What we know about 2026-XJ today: An asteroid striking Earth would be PERFECTLY INELASTIC — it becomes part of Earth (after vaporizing). A pure gravity flyby (no contact) is ELASTIC. DART was somewhere in between.

How today connects: Once you know which TYPE you have, you know which equations to use.

VOCAB · ELASTIC COLLISION

Definition: A collision where both MOMENTUM and KINETIC ENERGY are conserved. Objects bounce off without losing energy.

Example: Superball off concrete. Newton's cradle. Billiard balls (approximately). Gravity flybys.

Cognates: *Sp. colisión elástica · Pt. colisão elástica · HC kolizyon elastik*

Watch out: No real collision is 100% elastic — some energy always goes to heat/sound. But many problems treat them as elastic.

VOCAB · INELASTIC COLLISION

Definition: A collision where MOMENTUM is conserved but KINETIC ENERGY is NOT — some KE goes to heat, sound, deformation.

Example: Most car crashes. A dropped clay ball that flattens but doesn't stick. Bouncing tennis ball (slightly inelastic).

Cognates: *Sp. colisión inelástica · Pt. colisão inelástica · HC kolizyon inelastik*

Watch out: The momentum is STILL conserved. Only the kinetic ENERGY decreases. Don't say 'momentum is lost.'

VOCAB · PERFECTLY INELASTIC COLLISION

Definition: A collision where the objects STICK TOGETHER after impact. Momentum conserved; maximum KE lost.

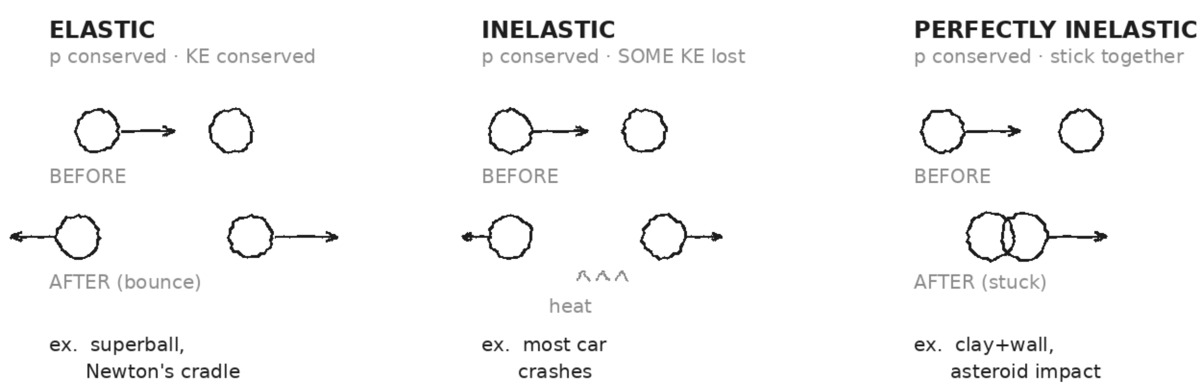
Example: Clay ball into wall. Bullet into wooden block (ballistic pendulum). Asteroid impact.

Cognates: *Sp. perfectamente inelástica · Pt. perfeitamente inelástica · HC perfekteman inelastik*

Watch out: 'Perfectly' is a technical word — it doesn't mean ideal/good. It means 'stuck together completely.'

DOODLE ZONE

Add ONE more real example to each column from your own life. Circle the ones you've seen.



CLASSIFICATION DRILL

Classify each as elastic (E), inelastic (I), or perfectly inelastic (PI). Be ready to defend.

CLASSIFY EACH

Two billiard balls bouncing off each other .... ____

A car rear-ends a stalled truck, bumpers lock .... ____

Football tackle — players grip each other and tumble .... ____

Bullet hits a wood block and sticks .... ____

Two magnets repel each other and bounce apart without touching .... ____

Asteroid hits Earth .... ____

Newton's cradle — end ball stops, opposite end ball moves .... ____

Wet clay ball thrown against a wall, sticks .... ____

⚠	ALL THREE types conserve momentum. Only ELASTIC conserves kinetic energy. Don't confuse 'energy lost' with 'momentum lost.' They're different rules.

EXIT TICKET — DAY 6

A meteorite buries itself in the desert. What kind of collision is this? Where did its kinetic energy go?

Today's Targets — Rate Yourself

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can define elastic, inelastic, and perfectly inelastic collisions.	○	○	○	☐
I can identify which conservation law(s) apply to each type.	○	○	○	☐
I can classify real-world collisions confidently.	○	○	○	☐

Day 7 — Vernier Collision Lab — Verify Conservation · Investigation 3.1

Today's Target

I can run elastic and perfectly inelastic cart collisions with photogates, compute Σp_{before} and Σp_{after}, and report the % difference with one credible source of error.

ASTEROID THREAD

What we know about 2026-XJ today: We've been STATING that Σp is conserved for 7 days. Today we MEASURE it.

How today connects: If our lab carts confirm conservation, we can trust the same rule for asteroids. Days 11–13 depend on it.

LAB SETUP — INVESTIGATION 3.1

Driving question: How well does Σp_before match Σp_after in our lab carts? And does it work for BOTH elastic and perfectly inelastic collisions?

Equipment:

- Vernier dynamics track (1 per pair) + 2 carts
- Magnetic bumpers on each cart (for the elastic trial)
- Velcro pads on each cart (for the perfectly inelastic trial)
- 2 photogates + LabQuest + Logger Pro or Graphical Analysis
- Balance to weigh each cart (record m_1 and m_2 first)

What you do:

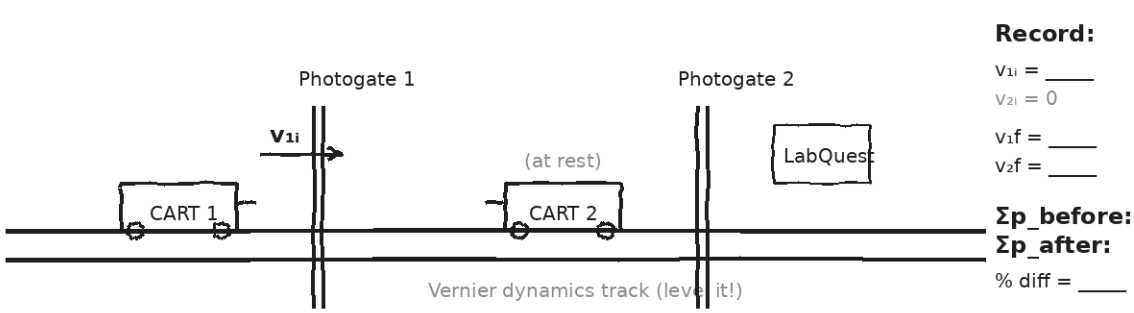
1. Level the track. Weigh both carts and record m_1, m_2.
2. ELASTIC trials (magnet bumpers): cart 2 at rest. Push cart 1 gently. Photogates read v_{1i}, v_1f, and v_2f.
3. Compute $\Sigma p\_before = m_1v_{1i}$ and $\Sigma p\_after = m_1v_1f + m_2v_2f$. Compare.
4. PERFECTLY INELASTIC trials (Velcro): cart 2 at rest. Push cart 1; carts stick.
5. Compute $\Sigma p\_after = (m_1+m_2)\cdot v\_combined$. Compare to Σp_before.
6. Two trials of each type. Compute % difference for each.

Safety: Cart speeds gentle. Photogates clamped firmly. Hands clear of cart paths.

How this lab serves the year's question: If conservation holds in carts within a few percent, the asteroid Days 11–13 stand.

DOODLE ZONE

Sketch your setup. Record m_1 and m_2 on the diagram. Fill v_before and v_after numbers as you go.



DATA — ELASTIC TRIALS (MAGNET BUMPERS, CART 2 AT REST)

Trial	v_{1i} (m/s)	v_{1f} (m/s)	v_{2f} (m/s)	Σp_before	Σp_after	% diff
1						
2						

DATA — PERFECTLY INELASTIC TRIALS (VELCRO, CART 2 AT REST)

Trial	v_{1i} (m/s)	v_combined (m/s)	Σp_before	Σp_after	% diff	Notes
1						
2						

OBSERVATION & ANALYSIS

In your trials, did Σp_after match Σp_before? Within what %?

If there's a difference, what physical thing leaked momentum out? Was it the same for elastic and inelastic trials?

My elastic Σp matched within __________ %; my perfectly inelastic Σp matched within __________ %. The biggest source of error was __________.

Asteroid connection: how does this serve the year's question?

Fill in the frame:

Conservation of momentum predicts Σp_after = Σp_before. My elastic trials confirmed this within ____ %; my inelastic trials within ____ %. The asteroid math on Days 11–13 is ____.

EXIT TICKET — DAY 7
Write a one-sentence CLAIM about what your lab showed, then one sentence of EVIDENCE (your numbers), then one sentence of REASONING (why the % difference is small).

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can run a Vernier two-cart elastic collision.	○	○	○	☐
I can run a Vernier two-cart perfectly inelastic collision.	○	○	○	☐
I can compute Σp before and after and report % difference.	○	○	○	☐
I can name a credible source of error.	○	○	○	☐

Day 8 — Inelastic Collisions — Stick-Together Worked Examples

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Today's Target

I can solve a 1D perfectly inelastic collision using $m_1v_1 + m_2v_2 = (m_1+m_2)v\_f$, with correct signs and units.

ASTEROID THREAD

What we know about 2026-XJ today: An asteroid impact is the planetary version of 'stick together.' Today's math is what we'd use to estimate Earth's velocity change from a 2026-XJ strike.

How today connects: Same equation. Different scale. The arithmetic generalizes.

VOCAB · BALLISTIC PENDULUM

Definition: A classic stick-together problem: a small fast object embeds in a larger object at rest. Use conservation of momentum.

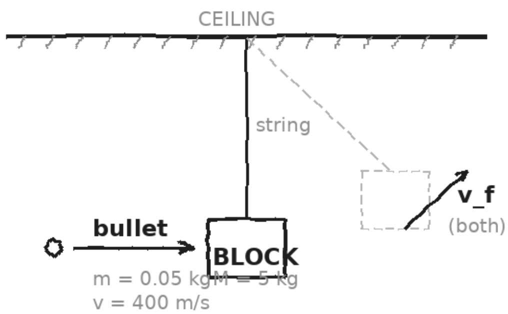
Example: A 0.05-kg bullet at 400 m/s into a 5-kg block at rest. After impact: (bullet + block) move together at $v\_f \approx 4$ m/s.

Cognates: *Sp. péndulo balístico · Pt. pêndulo balístico*

Watch out: Don't try $\frac{1}{2}mv^2 = \frac{1}{2}(M+m)v\_f^2$ — that's kinetic energy, which IS NOT conserved here. Use momentum: $mv = (M+m)v\_f$.

DOODLE ZONE

Use Σp to find v_f. Direction same as bullet. Then circle which conservation law you used: momentum / energy. (Hint: only ONE works here.)



Step 1 — momentum:

m·v + 0 = (m+M)·v_f

Solve for v_f.

v_f = ___________ m/s

Step 2 — direction?

Same as bullet:

_______ ward.

(Energy LOST is huge — Unit 4.)

WORKED EXAMPLE — BALLISTIC PENDULUM

A 0.05-kg bullet moving east at 400 m/s embeds itself in a 5-kg block at rest. Find the velocity of the bullet+block immediately after impact.

GIVEN	$m = 0.05$ kg · $v = +400$ m/s · $M = 5$ kg · $V = 0$
EQUATION	$m \cdot v + M \cdot V = (m + M) \cdot v_f$
WORK	$(0.05)(400) + (5)(0) = (0.05 + 5) \cdot v_f$ $20 = 5.05 \cdot v_f$ $v_f = 20/5.05 \approx 3.96$ m/s
ANSWER	$v_f \approx 3.96$ m/s east. The block + bullet move together at about 4 m/s.

WORKED EXAMPLE — HEAD-ON CAR COLLISION (FORCE SIGNS)

A 1200-kg car going east at 25 m/s collides head-on with an 1800-kg car going west at 15 m/s. They lock together. Find the final velocity.

GIVEN	$m_1 = 1200$ kg, $v_1 = +25$ m/s (east) $m_2 = 1800$ kg, $v_2 = -15$ m/s (west)
EQUATION	$m_1v_1 + m_2v_2 = (m_1 + m_2)\cdot v_f$
WORK	$(1200)(25) + (1800)(-15) = (3000)\cdot v_f$ $30{,}000 + (-27{,}000) = 3000\cdot v_f$ $3{,}000 = 3000\cdot v_f$ $v_f = +1.0$ m/s
ANSWER	+1.0 m/s — the wreckage drifts EAST slowly. The heavier westbound car lost out narrowly.

YOUR TURN — REAR-END COLLISION

A 1000-kg car going east at 20 m/s rear-ends a 2000-kg stalled truck. They lock together. Find the final velocity.

GIVEN	
EQUATION	
WORK	
ANSWER	

⚠	ALWAYS write the sign convention before you compute. East = + (or whatever you pick). Then assign signs to every v BEFORE you sum.

EXIT TICKET — DAY 8
A 1000-kg car at 20 m/s east rear-ends a 2000-kg stalled truck. They lock. What's the final velocity?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)*   **2 = Almost** *(I can do this with some help or some errors)*   **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	**1**	**2**	**3**	**Help?**
I can write the perfectly-inelastic equation correctly.	○	○	○	☐
I can assign signs to opposite-direction velocities.	○	○	○	☐
I can solve a ballistic-pendulum-style problem.	○	○	○	☐

Day 9 — Elastic Collisions — Bounce-Apart Worked Examples

Today's Target

I can solve a 1D elastic collision using conservation of momentum AND conservation of kinetic energy, and I can apply the equal-mass and one-at-rest simplifications.

ASTEROID THREAD

What we know about 2026-XJ today: A gravity flyby (no contact) is essentially elastic — both p and KE conserved.

How today connects: The math here is what we'd use to predict an asteroid's path AFTER a gravity-assist swing-by.

VOCAB · ELASTIC — ONE AT REST

Definition: Special case: object 1 (mass m_1, speed v_{1i}) strikes object 2 (mass m_2, at rest). Result: v_1f and v_2f from simple formulas.

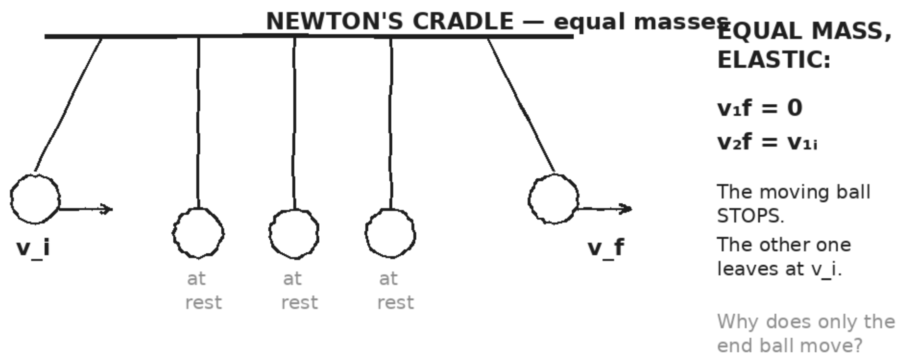
Example: $v_1f = ((m_1-m_2)/(m_1+m_2))\cdot v_{1i}$; $v_2f = (2m_1/(m_1+m_2))\cdot v_{1i}$. Both directly from the two conservation rules.

Cognates: *Sp. (caso especial) un cuerpo en reposo · Pt. um corpo em repouso*

Watch out: If $m_1 = m_2$: $v_1f = 0$, $v_2f = v_{1i}$. The mover STOPS, the resting one takes off (Newton's cradle).

DOODLE ZONE

Why does only the END ball move? Use the equal-mass formula to predict v_1f and v_2f. What does the formula say?



EQUATION · ANATOMY — THE ELASTIC PAIR

p: $m_1v_{1i} + m_2v_{2i} = m_1v_1f + m_2v_2f$

KE: $\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_1f^2 + \frac{1}{2}m_2v_2f^2$

Two equations. Two unknowns (v_1f, v_2f). Solvable. When m_2 is at rest, use the simplified formulas in the vocab box above.

WORKED EXAMPLE — 0.5 KG HITS 1.0 KG AT REST

A 0.5-kg ball at 4 m/s east strikes a 1.0-kg ball at rest, elastically. Find v_1f and v_2f.

GIVEN	$m_1 = 0.5$ kg, $v_{1i} = +4$ m/s · $m_2 = 1.0$ kg, $v_{2i} = 0$
EQUATION	$v_1f = ((m_1-m_2)/(m_1+m_2))\cdot v_{1i}$; $v_2f = (2m_1/(m_1+m_2))\cdot v_{1i}$
WORK	$v_1f = ((0.5 - 1.0)/(0.5 + 1.0))\cdot(4) = (-0.5/1.5)\cdot 4 \approx -1.33$ m/s $v_2f = (2\cdot 0.5/1.5)\cdot(4) = (1/1.5)\cdot 4 \approx 2.67$ m/s
ANSWER	$v_1f \approx -1.33$ m/s (bounces BACK west). $v_2f \approx 2.67$ m/s east.

PATTERN-FINDING

Use the simplified one-at-rest formulas to predict each case. Sketch a quick before/after to check your intuition.

PREDICT EACH
(a) $m_1 = m_2$ (equal masses): v_{1f} = _______, v_{2f} = _______ (b) $m_1 \gg m_2$ (big hits small): $v_{1f} \approx$ _______ (continues forward, almost unchanged?), $v_{2f} \approx$ _______ (c) $m_1 \ll m_2$ (small hits big): $v_{1f} \approx$ _______ (bounces back?), $v_{2f} \approx$ _______ (barely moves?)

YOUR TURN — EQUAL-MASS SUPERBALLS

A 0.2-kg superball moving east at 5 m/s strikes a 0.2-kg superball at rest, elastically. Use the equal-mass result to predict v_{1f} and v_{2f}.

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Newton's cradle uses the equal-mass rule. The moving ball stops; an identical ball on the far end leaves at the original speed. Conservation of both p AND KE forces it.

EXIT TICKET — DAY 9
A 0.2-kg superball at 5 m/s east strikes a 0.2-kg ball at rest, elastically. What happens to each?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can apply the one-at-rest simplified formulas.	○	○	○	☐
I can predict equal-mass / big-hits-small / small-hits-big patterns.	○	○	○	☐
I can solve a 1D elastic collision.	○	○	○	☐

Day 10 — 2D Collisions — Glancing Blows

Today's Target

I can apply conservation of momentum SEPARATELY to the x- and y-components, and I can solve a 2D glancing collision.

ASTEROID THREAD

What we know about 2026-XJ today: An asteroid that's nudged doesn't just slow down — it DEFLECTS sideways.

How today connects: 2D momentum is the framework for asking 'how much sideways?' Tomorrow's DART analysis depends on it.

VOCAB · COMPONENT-WISE CONSERVATION

Definition: Total momentum is a VECTOR. Conservation applies separately to its x- and y-components: $\Sigma p_{x,before} = \Sigma p_{x,after}$ AND $\Sigma p_{y,before} = \Sigma p_{y,after}$.

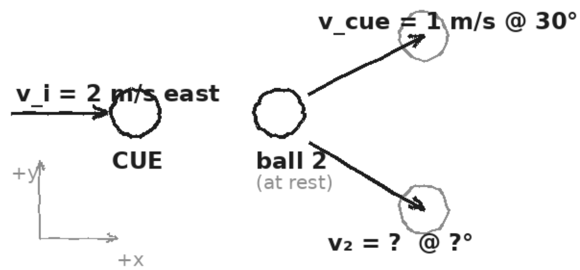
Example: A cue ball east strikes a stationary ball; afterward both move at angles. Σp_x and Σp_y each balance.

Cognates: *Sp. componentes (x, y) · Pt. componentes (x, y)*

Watch out: ONE conservation equation isn't enough in 2D. Use TWO (one per direction). They give two equations for the two unknowns.

DOODLE ZONE

Use the equations on the right to find v_2 and θ. Hint: equal masses cancel out.



Σp_x: m·v_i = m·v_cue·cos30° + m·v_2·cos θ
(masses cancel)

Σp_y: 0 = m·v_cue·sin30° + m·v_2·sin θ
(one negative — opposite y dirs)

Two equations, two unknowns:
v_2 = ___________ m/s
θ = ___________°

Conservation works
x AND y separately.

WORKED EXAMPLE — BILLIARDS GLANCING COLLISION

A 0.16-kg cue ball moves east at 2 m/s and strikes a 0.16-kg ball at rest. Afterward, the cue ball moves at 1 m/s, 30° north of east. Find the second ball's speed and direction.

GIVEN	$m_1 = m_2 = 0.16$ kg (cancels) · v_cue_i = +2 m/s east · v_cue_f = 1 m/s @ +30° · v_2_i = 0
EQUATION	Σp_x: m·v_i = m·v_cue_f·cos30° + m·v_2·cos θ Σp_y: 0 = m·v_cue_f·sin30° − m·v_2·sin θ
WORK	Drop the masses (cancel): x: $2 = 1\cdot\cos 30° + v_2\cdot\cos\theta \rightarrow v_2\cdot\cos\theta = 2 - 0.866 = 1.134$ y: $0 = 1\cdot\sin 30° - v_2\cdot\sin\theta \rightarrow v_2\cdot\sin\theta = 0.500$ $\tan\theta = 0.500/1.134 \rightarrow \theta \approx 23.8°$ (south of east) $v_2 = 0.500/\sin 23.8° \approx 1.24$ m/s
ANSWER	Ball 2: ≈ 1.24 m/s at ≈ 23.8° south of east.

YOUR TURN — DIFFERENT ANGLE

Same setup, but now the cue ball goes 1.5 m/s at 45° north of east after impact. Find the second ball's speed and direction.

GIVEN	
EQUATION	
WORK	
ANSWER	

⚠	The y-components must SUM TO ZERO if the cue ball was originally moving only along x. If one ball goes 'north', the other must go 'south' (in the y-component).

Fill in the frame:

We can't just balance the total momentum magnitudes. We need to balance x and y ____, because momentum is a ____.

EXIT TICKET — DAY 10
Why must p_x and p_y each be conserved separately, not just total p? Use the word 'vector' in your answer.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can split momentum into x and y components.	○	○	○	☐
I can apply conservation to each axis independently.	○	○	○	☐
I can solve a 2D glancing collision for the unknown ball's velocity.	○	○	○	☐

Day 11 — DART Case Study — Nudging an Asteroid

Today's Target

I can use conservation of momentum to compute the Δv DART gave to Dimorphos, and I can apply the same calculation to 2026-XJ to estimate what it would take to deflect it.

ASTEROID THREAD

What we know about 2026-XJ today: On Sept 26, 2022, NASA's DART spacecraft (~600 kg, ~6 km/s) crashed into a small asteroid called Dimorphos ($\sim 5 \times 10^9$ kg). It changed Dimorphos's orbital period by 32 minutes.

How today connects: This is the day the unit pays off. Today's math is the math we'd use on 2026-XJ. If the Δv we need is DART-scale, deflection is on the table. If not, we'd need a much bigger push (or a different strategy).

VOCAB · DART (DOUBLE ASTEROID REDIRECTION TEST)

Definition: NASA's 2022 mission that deliberately crashed a 600-kg spacecraft into a small asteroid to test whether kinetic impact can change its trajectory.

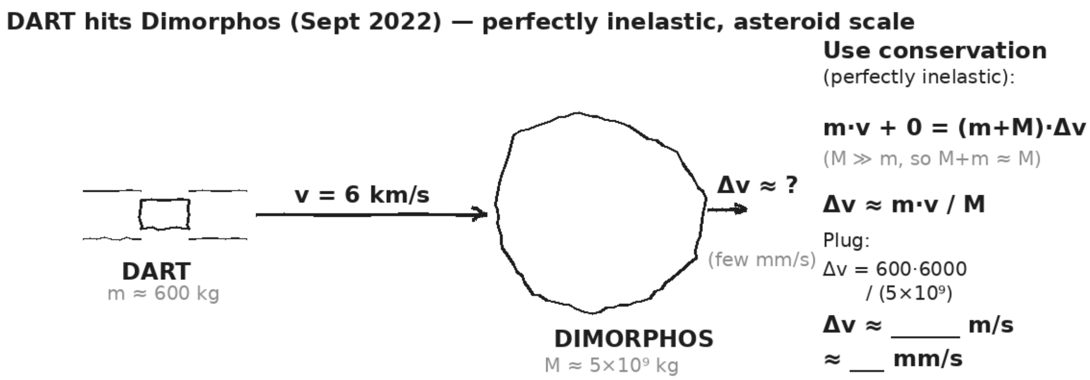
Example: DART hit Dimorphos at ~6 km/s, delivering only ~3 mm/s of Δv to the asteroid — but that tiny push, accumulated over time, shifts its position by km and km.

Cognates: *Sp. (misión DART) · Pt. (missão DART) · HC misyon DART*

Watch out: The Δv from DART is TINY (mm/s). But that tiny shift, over years of orbital motion, produces a measurable position change. The 'small Δv × long time' equation is the key.

DOODLE ZONE

Plug the DART numbers into Δv = m·v / M. Get the answer in m/s, then convert to mm/s. Tiny — but enough to matter.



WORKED EXAMPLE — WHAT Δv DID DART GIVE DIMORPHOS?

DART ($m = 600$ kg) hit Dimorphos ($M = 5 \times 10^9$ kg) at $v = 6 \times 10^3$ m/s, treating the impact as perfectly inelastic. Find the velocity change Δv of the asteroid. (Hint: $M \gg m$, so $(m+M) \approx M$.)

GIVEN	$m = 600$ kg · $v = 6000$ m/s · $M = 5\times10^9$ kg · M_asteroid initially at rest in DART's frame
EQUATION	$m \cdot v + M \cdot 0 = (m + M)\cdot\Delta v \approx M\cdot\Delta v$ (since $M \gg m$) $\Delta v \approx m \cdot v / M$
WORK	$\Delta v \approx (600 \times 6000) / (5 \times 10^9)$ $\Delta v \approx 3.6 \times 10^6 / 5 \times 10^9$ $\Delta v \approx 7.2 \times 10^{-4}$ m/s ≈ 0.72 mm/s (NASA measured ≈ 2.7 mm/s with momentum enhancement from ejecta — same order of magnitude.)
ANSWER	≈ 0.72 mm/s (less than a millimeter per second). Tiny — but it adds up.

WHY DOES A TINY Δv MATTER?

$\Delta v \times$ time = position offset

If $\Delta v \approx 1$ mm/s and we have 1 year (3.15×10^7 s) of lead time:

offset = $(10^{-3}$ m/s$) \times (3.15 \times 10^7$ s$) \approx 3.15 \times 10^4$ m ≈ 31.5 km

More lead time = more offset. More Δv = more offset. Linear in both.

YOUR TURN — WHAT ΔV TO DEFLECT 2026-XJ BY AN EARTH-RADIUS?

2026-XJ has mass $M \approx 1 \times 10^9$ kg and will pass Earth in 6 months ($t \approx 1.5 \times 10^7$ s). We need to shift its position by $R_{Earth} \approx 6.4 \times 10^6$ m to miss us. What Δv does it need? Then: would a DART-scale strike (say a 600-kg, 6-km/s spacecraft) deliver it?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	When you compute the required Δv and compare to what a single DART would give: the gap tells you whether one impactor is enough, or whether we'd need many — or sooner.

Fill in the frame:

To deflect 2026-XJ by one Earth-radius in 6 months, we need $\Delta v \approx$ ____. A single DART-class strike would give about ____.

So we'd need ____ DARTs (or more lead time).

EXIT TICKET — DAY 11
Why doesn't NASA need a HUGE Δv to deflect an asteroid? Use the words 'lead time' and 'small change in velocity' in your answer.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can apply $\Sigma p_{before} = \Sigma p_{after}$ to a DART-style impact.	○	○	○	☐
I can compute the required Δv from a desired position shift and lead time.	○	○	○	☐
I can compare a needed Δv to what a real spacecraft could deliver.	○	○	○	☐

Day 12 — Asteroid Impact — What Momentum Says About Damage

Today's Target

I can compute the average force of an asteroid impact using $F = \Delta p/\Delta t$ and explain why Earth's velocity changes negligibly even though the impact is catastrophic locally.

ASTEROID THREAD

What we know about 2026-XJ today: If deflection fails, momentum gives our FIRST picture of the impact. Earth gains almost no velocity (we outmass the asteroid 10^{15} to 1). But the asteroid delivers all of its momentum in 0.1 s.

How today connects: Today is the OTHER branch of the unit. Yesterday: deflect. Today: what if we don't?

VOCAB · LOCAL VS. GLOBAL MOMENTUM

Definition: In a planetary collision, the asteroid's momentum is transferred to a SMALL local region of Earth in a SHORT time. Earth as a whole barely moves.

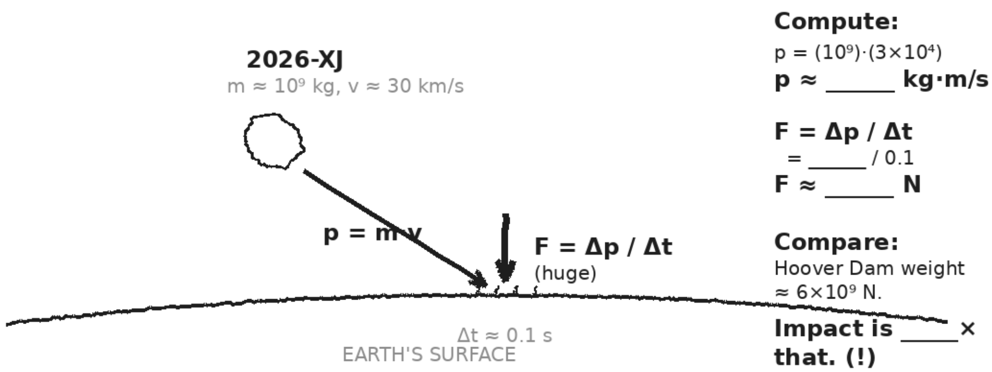
Example: 2026-XJ at 30 km/s carries $p \approx 3\times10^{13}$ kg·m/s. Earth (6×10^{24} kg) would gain $\Delta v \approx 5\times10^{-12}$ m/s — utterly undetectable. The impact site, though, absorbs the entire force.

Cognates: *Sp. local vs. global · Pt. local vs. global*

Watch out: Conservation says Earth's CENTER-OF-MASS velocity barely changes. But the LOCAL force at the impact site is enormous. Both are true at once.

DOODLE ZONE

Compute p, then F = p / 0.1 s. Compare F to the weight of Hoover Dam (≈ 6×10^9 N). How many Hoover Dams of force?



WORKED EXAMPLE — 2026-XJ IMPACT FORCE

If 2026-XJ ($m = 1 \times 10^9$ kg) hits Earth at $v = 3 \times 10^4$ m/s and stops in $\Delta t \approx 0.1$ s, what is the AVERAGE force during impact?

GIVEN	$m = 1 \times 10^9$ kg · $v = 3 \times 10^4$ m/s · $v_f = 0$ · $\Delta t = 0.1$ s
EQUATION	$\Delta p = m\cdot(v_f - v_i) = -m\cdot v_i$; $F = \Delta p / \Delta t$
WORK	$\Delta p = -(1 \times 10^9)(3 \times 10^4) = -3 \times 10^{13}$ kg·m/s $F = -3 \times 10^{13} / 0.1 = -3 \times 10^{14}$ N
ANSWER	≈ 3×10^{14} N of force. About 50,000 × the weight of Hoover Dam, applied in a tenth of a second.

WORKED EXAMPLE — Earth's velocity change (almost zero)

Using conservation of momentum, find Earth's velocity change after a 2026-XJ impact. ($M\_Earth = 5.97 \times 10^{24}$ kg.)

GIVEN	$p\_asteroid = 3 \times 10^{13}$ kg·m/s · $M\_Earth = 5.97 \times 10^{24}$ kg
EQUATION	$p\_asteroid + 0 = (M\_Earth) \cdot \Delta v\_Earth$ (asteroid essentially absorbed) $\Delta v\_Earth = p\_asteroid / M\_Earth$
WORK	$\Delta v\_Earth = (3 \times 10^{13}) / (5.97 \times 10^{24})$ $\Delta v\_Earth \approx 5 \times 10^{-12}$ m/s
ANSWER	$\approx 5 \times 10^{-12}$ m/s — utterly undetectable. Earth doesn't 'move.' But the impact site catches the full force.

⚠	Momentum gives us the FORCE. It does NOT capture the energy (heat, shockwave, crater). That comes in Unit 4. Today's number is one PART of the disaster picture.

Fill in the frame:

Earth's velocity barely changes after impact because Earth's

____ is roughly 10^{15} times the

asteroid's. But the LOCAL force at the impact site is enormous because

____ is delivered in a very short

____.

EXIT TICKET — DAY 12

Why is Earth's velocity change after an asteroid impact essentially zero, even though the impact is catastrophic locally?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	**1**	**2**	**3**	**Help?**
I can compute the AVERAGE force of an asteroid impact using $F = \Delta p/\Delta t$.	○	○	○	☐
I can compute Earth's negligible Δv from a conservation argument.	○	○	○	☐
I can explain why a tiny Earth Δv coexists with a catastrophic local force.	○	○	○	☐

Day 13 — Two-Branch Analysis — Deflect or Impact?

Today's Target

I can run a full 2026-XJ analysis on both branches (deflection and impact) and articulate which two ideas from the unit were load-bearing for the conclusion.

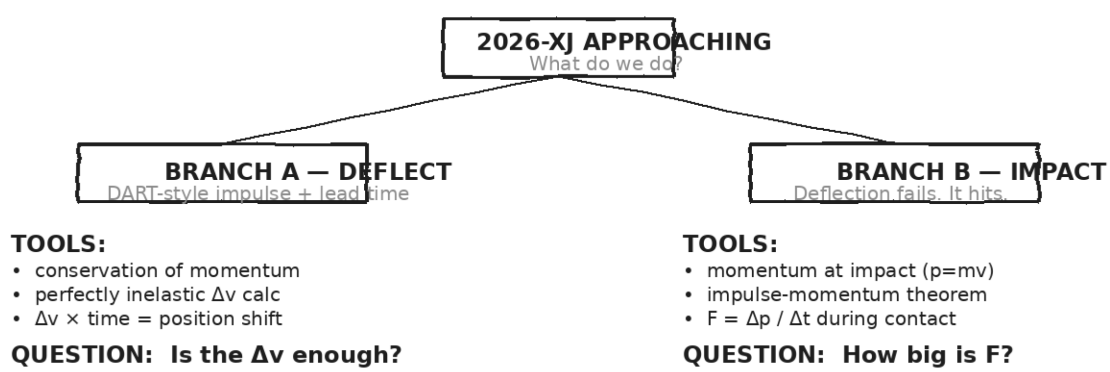
ASTEROID THREAD

What we know about 2026-XJ today: We have every tool we need. Days 11 and 12 each gave us a branch. Today the class does both, compiles, and reports.

How today connects: Tomorrow is the transfer task. Today is the rehearsal — done in groups, with talk and revision.

DOODLE ZONE

On the LEFT branch, list the physics tools you'll use. On the RIGHT branch, do the same. Then circle one tool that shows up in BOTH.



STATION 1 — DEFLECTION (ASTEROID + DART-STYLE IMPACTOR)

Given: 2026-XJ has $M = 1 \times 10^9$ kg, will pass Earth in $t = 1.5 \times 10^7$ s (6 months). We need to shift its position by 6.4×10^6 m (one Earth-radius). A DART-class impactor has $m = 600$ kg at $v = 6 \times 10^3$ m/s. (a) Find the Δv from ONE DART-class impactor. (b) Find the Δv we NEED (using Δv = shift / time). (c) How many DART impactors would it take?

GIVEN	
EQUATION	
WORK	
ANSWER	

STATION 2 — IMPACT (WHAT WE'D LIVE THROUGH)

Given: 2026-XJ at $v = 3 \times 10^4$ m/s, $M = 1 \times 10^9$ kg, hits Earth and stops in $\Delta t = 0.1$ s. (a) Find the asteroid's momentum at impact. (b) Find the average impact force. (c) Compare F to: Hoover Dam weight (6×10^9 N), Burj Khalifa weight (5×10^9 N), or an earthquake's typical peak force.

GIVEN	
EQUATION	
WORK	
ANSWER	

CLASS CONSENSUS — WHAT DO WE TELL NASA?

After both stations, compare answers across groups. Write a 2-sentence consensus statement about (a) whether DART-scale deflection is realistic and (b) what the impact branch would mean.

OUR CLASS CONSENSUS

★	Real planetary defense analysts do exactly this kind of two-branch reasoning. Your numbers are within an order of magnitude of NASA's. The fictional asteroid; the math is real.

Fill in the frame:

Of all the ideas from Unit 3, the TWO most load-bearing for today's analysis were ____ (used in deflection) and ____ (used in impact). Both rely on conservation of ____.

EXIT TICKET — DAY 13
Of all the things you learned in Unit 3, which TWO ideas were load-bearing for today's analysis? Why?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	**1**	**2**	**3**	**Help?**
I can run the deflection-branch calculation start to finish.	○	○	○	☐
I can run the impact-branch calculation start to finish.	○	○	○	☐
I can articulate which two unit ideas were load-bearing for the conclusion.	○	○	○	☐

Day 14 — Unit 3 Transfer Task — Assessment Day

ASTEROID THREAD

What we know about 2026-XJ today: Every tool from Unit 3 is in your hands: $p = mv$, $J = F\Delta t = \Delta p$, conservation, three collision types, 2D components, and the DART/impact branches.

How today connects: One of the 4 transfer problems anchors in 2026-XJ — given a DART-style Δv, compute the lead time required for an Earth-radius lateral shift.

TRANSFER TASK — UNIT 3 · DAY 14

Today you receive the Unit 3 Transfer Task on a separate sheet. It is 4 multi-part problems. You will work independently for ~40 minutes.

- Problem 1 — Momentum + impulse. A baseball-bat collision: given m, v_i, v_f, and contact time, find Δp and the average force.
- Problem 2 — Ballistic pendulum (perfectly inelastic). Given bullet mass, bullet speed, and block mass, find the combined velocity. Then find the kinetic energy lost (preview to Unit 4).
- Problem 3 — 2D billiards. Given m_1, m_2, v_{cue_i}, and the cue ball's final speed + angle, find the second ball's final speed and direction.
- Problem 4 — Asteroid deflection. Given the Δv from a DART-style impactor and a target lateral shift of 6,400 km, how much lead time is needed?

HOW THIS WILL BE SCORED — 4 DIMENSIONS, EACH 0–4

Dimension	What a strong (3–4) response shows
Science	Correct physics: vectors, kinematics, units.
Reasoning	Sound method; uncertainty explained, not just stated.
Communication	Work shown, units labeled, organized, readable.
Transfer-to-Asteroid	Honest public-facing connection to 2026-XJ.

MY WORK — DAY 14

AFTER THE TASK — FINAL UNIT MARZANO (RATE EACH UNIT 3 TARGET)

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet *(I can't do this without help)* **2 = Almost** *(I can do this with some help or some errors)* **3 = Got It** *(I can do this independently and explain it)*

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
K1 — Momentum as a vector p = m·v.	○	○	○	☐
K2 — Impulse-momentum theorem J = F·Δt = Δp.	○	○	○	☐
K3 — Conservation of momentum and its condition.	○	○	○	☐
K4 — Elastic vs. inelastic vs. perfectly inelastic.	○	○	○	☐
S1 — Compute momentum, Δp, and impulse.	○	○	○	☐
S2 — Solve a perfectly inelastic 1D problem.	○	○	○	☐
S3 — Solve a 1D elastic problem (general + special cases).	○	○	○	☐
S4 — Run a Vernier collision experiment and report % difference.	○	○	○	☐
R1 — Reason about how Δt trades off with F for the same Δp.	○	○	○	☐
R2 — Apply 2D conservation to glancing collisions.	○	○	○	☐
R3 — Reason about asteroid deflection using Δv × time.	○	○	○	☐

Equation Reference — Unit 3

Keep this page open while you work. Every equation below is on the MCAS Equation Sheet — you do NOT need to memorize them. You DO need to recognize when each one applies.

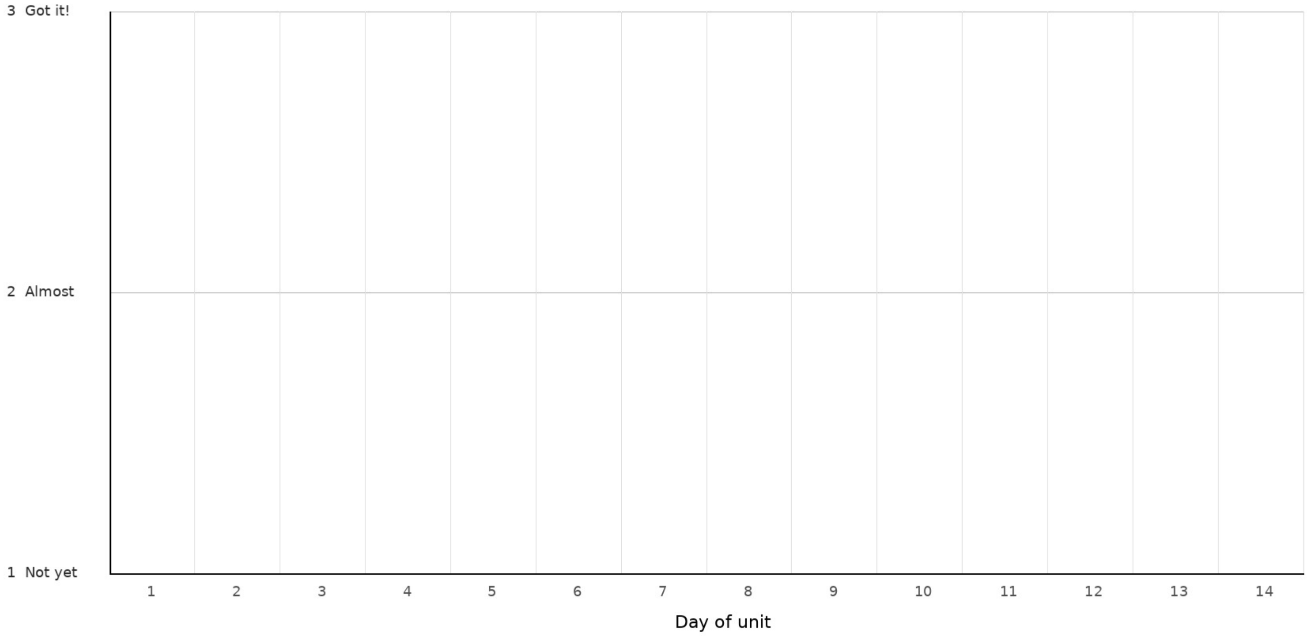
Concept	Equation	When to use it
Momentum	$p = m \cdot v$	Units kg·m/s. Vector — same direction as v. Big mass × big speed = big p.
Impulse-momentum	$J = F \cdot \Delta t = \Delta p$	Units N·s = kg·m/s. The bridge from FORCE (Unit 1) to MOMENTUM. Same Δp can come from big F·short Δt OR small F·long Δt.
Conservation of momentum	$\Sigma p\_before = \Sigma p\_after$	Holds when no NET external force acts on the system. Applies in EVERY collision type.
Perfectly inelastic (stick)	$m_1v_1 + m_2v_2 = (m_1+m_2)v\_f$	Objects stick together after impact. ONE unknown, v_f. Asteroid impact and ballistic pendulum are this case.
Elastic — general	Σp and ΣKE both conserved	Two equations: $m_1v_{1i} + m_2v_{2i} = m_1v_1f + m_2v_2f$ AND $\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_1f^2 + \frac{1}{2}m_2v_2f^2$. No KE lost. Bouncing superball, Newton's cradle, gravity flyby.
Elastic — m_2 at rest	$v_1f = ((m_1-m_2)/(m_1+m_2)) \cdot v_{1i}$ $v_2f = (2m_1/(m_1+m_2)) \cdot v_{1i}$	Special case shortcut. Predict equal-mass ($v_1f=0$, $v_2f=v_{1i}$ — Newton's cradle), small hits big (bounces back), big hits small.
2D conservation	$\Sigma p\_x\_before = \Sigma p\_x\_after$ $\Sigma p\_y\_before = \Sigma p\_y\_after$	Vector momentum splits into COMPONENTS. Conservation applies to x and y SEPARATELY. Two equations, often two unknowns.
Asteroid deflection	$\Delta v \approx m\_DART \cdot v\_DART / M\_asteroid$	From conservation, perfectly inelastic, M ≫ m approximation. Tiny per-strike Δv accumulates over years.

Constants and reference quantities:

$M\_Earth = 5.97 \times 10^{24}$ kg · $R\_Earth = 6.37 \times 10^{6}$ m (≈ 6371 km) · Earth orbital speed ≈ 3.0×10^{4} m/s · DART: m ≈ 600 kg, v ≈ 6 km/s · Dimorphos M ≈ 5×10^{9} kg · 2026-XJ m ≈ 10^{9} kg, v ≈ 3×10^{4} m/s

Mastery Growth Chart — Unit 3

Each day, transfer your Marzano 1-2-3 rating onto the chart. Use a different mark or color for each target you track. By the end of the unit you'll have a visible record of your growth.



LEGEND — WHICH MARK / COLOR = WHICH TARGET
Mark: ______ Target: ______________________
Mark: ______ Target: ______________________
Mark: ______ Target: ______________________
NOTES — END-OF-UNIT REFLECTION